

OSCAR MOXON

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EDUCATION

2023 – Present	MSc. in Artificial Intelligence, King's College London, UK (1:1) Modules: <i>Pattern Recognition and Deep Learning, Data Mining, Agents and Multi-Agent Systems, AI Planning, Computer Vision, Optimisation Methods, Machine Learning</i>
2020 – 2023	BSc. in Economics and Management Sciences, University of Southampton, UK First Class 1:1 Recipient of Dean's Award 2022 & 2023 for Outstanding Attainment, mentioned in Graduate Honours Modules: <i>Applied Macroeconomics, Risk Management, Industrial Economics, International Trade, Development Economics, Mathematics for Economics</i>

ACHIEVEMENTS

Global Top 10 Portfolio (of 980 teams), Bloomberg Trading Challenge (BTC)	Fall 2022
(x2) Deans Award for Outstanding Attainment	Academic Yr. 2020 - 2023
StabilityAI Researcher Sponsorship – CogX Conference	Fall 2023
10x Downing Street AI Hackathon: Winner	Spring 2024
3-Day-Startup, Barclays Eagle Labs: Winner	2021 & 2022
ECSS Pico Hackathon: Overall Winner	Summer 2022
Anthropic Hackathon: Nominated 2023	Winter 2023

PROFESSIONAL EXPERIENCE

Software Engineer, Bselected, London May 2023 – Present - Developing human-on-loop software for automated CV processing and deployed full-stack app through Azure Cloud. - Enabled 5-10x reduction in labour required to process a CV, with a 20% quality improvement using LLM tooling.
Innovative Technology Intern, BAE Systems, London Easter 2023 Researched internal systems and developed proposal for centralised data infrastructure to improve decision making latency.
Research Student, QuantX, Southampton May – Nov 2022 Automated manual intraday trading through volatility analysis. Developed a program to rank indicator performance on time-series data and generate heatmaps for promising strategies. Culminated in Bloomberg Challenge Top 10 finish.
Summer Intern, Level39 Accelerator, London Summer 2022 Involved in onboarding investors and startups. Shadowed data scientists in early-stage AI ventures, learning to implement recommenders with PyTorch. Organized discussions with founders in the fintech incubator. Managed invoicing and rent accounts.
Summer Placement, Canary Wharf Group, London Summer 2018 & 2022 Worked under board of directors incl. Head Architect and President Sir Iacobescu. Performed legal skyscraper "break-ins" to test security measures with Estate CyberSecurity Team. Produced reports for CFO on M&A and contributed to architectural projects.

RESEARCH

"Multi-Agent Debate Simulation," Research Thesis Advisor – Dr. Yulan He (Turing Institute) - Developing a multi-agent language system to augment the capability of collective agent swarms. - Exploring the limitations of language agents to promote novel scientific research and inquiry argumentation dialogues.
O. J. Moxon, <i>"The Threat of Automation: An Economic Perspective," Dissertation Literature Review, Mar 16, 2023. [Online]</i> . - Performed an assessment of methods of predicting economic impact of Machine Learning on Full Automation of Labour. - Proposed a general model for evaluating 'risk of automation' to furnish policymakers with more precise tools for navigating the capital benefits of technological change, ensuring equitable and effective redistribution strategies. 87%.

SKILLS

Programming	Fluency: Python Familiarity: C++, JavaScript, HTML, CSS, SQL, STATA, Ruby, Rust
Tools	Git, LaTeX, Scikit-Learn, Matplotlib, Pandas, NumPy, fastAI, PyTorch, Keras, MATLAB, OpenCV, XGBoost, Kaggle, TensorBoard, Hugging Face Transformers, Weka, Jekyll, Svelte, Docker, Flask, React
Mathematics	Statistics, Calculus, Linear Algebra, Game Theory, Coding Theory, Statistical Learning, Optimisation Methods
Algorithmics	Various Search and Ranking Strategies (Greedy, Heuristic, Genetic Algorithms, A* and WA* Search, Recommenders). Learning methods (SVM, Random Forest, Classification, Regression, Clustering, PCA, Boosting, Eval and Validate).
Models	Transformer Models, Generative Adversarial Networks, Policy Iteration and Q-Learning, RNNs, CNNs.

PROJECTS

No°10 Downing Street Generative AI Suite (MiDAS)

Apr 2024

- Employed state-of-the-art embeddings developed in the natural language domain and adapted it for search among government correspondence. Automated triaging using semantic analysis and long-context reasoning systems.
- Built out interface using full-stack skills, presented and won the big to present at 10 Downing Street to ministers.

Q-Learning and Policy RL Agent

Dec 23 – Feb 24

- Using Model-Free approaches with Q-learning, and Policy Iteration and Bayesian Optimisation.
- Very high performance and accuracy; achieved 'Legendary' class on relevant benchmarks.

WebVoyager Agent

Feb 2024

- Paper implementation and augmentation, using langchain to autonomously navigates the web like a human, performing tasks without specific APIs. Uses visual cues and intelligent decisions for web scraping, information retrieval and documentation.

High Performance Visual Classification and Segmentation for Text

Nov 2023

- Edge detection and image segmentation techniques. Achieved human-level performance with naïve methods on small dataset.
- Used batch normalisation, dropout, softmax activation, whitening.

Data Extraction (WOLLO)

Nov 22 – Jun 23

- Trained a learning algorithm using layout data inputs to process email data and reduce the use of non-recyclable paper receipts, before switching to a transformer-based visual processing stack.
- Building the backend AI for a mobile platform that classified user receipt data and offers budgeting and profit-sharing features for users. Involved designing, building, and presenting the Wollo application.
- Member of the 2023 FutureWorlds Cohort and a winner of the 2022 Barclays Eagle Labs Startup Competition, with the mission to democratise profit from users' online footprints.

Trading Algorithm

Nov 22

- Involved in research project focused using fundamental analysis to develop a high-performance, well hedged portfolio that generated returns of 18% during a month-long period buying security corrections.
- Developed a program to rank indicator performance on time series data.
- Team placed in the top 10 worldwide (and top 5 in Europe) against 980 university teams worldwide.

Diabolical Trumps

Aug 21

- Designed and conceptualised game of cards. Crowdfunded 4-figure seed (Kickstarter), manufactured game in Italy. Sold globally.

OTHER QUALIFICATIONS

Beechen Cliff School and Sixth Form, Bath

Sep 13 – Jul 19

- **A Levels:** Economics (A*), Physics (A), Mathematics (A), EPQ on 'Disrupting Monopolies' (A*)
- **GCSEs:** 9 (A - A*/9) and **Distinction*** for HPQ on 'Graphical Storytelling'.

Foundations of Machine Learning, Southampton University, 2021 [Surveyed]

Bloomberg Market Concepts, Bloomberg, 2021

Mathematics for Machine Learning, Coursera

Deep Learning Series, UCL X DeepMind

Interests: Have received 1.9m reads on my answers on Quora.com, most notably about figures such as Elon Musk and Renzo Piano. Publishing essays at <https://oscarmoxon.com>. Also published winner of a short story competition. Enjoy triathlons, hockey, and squash.